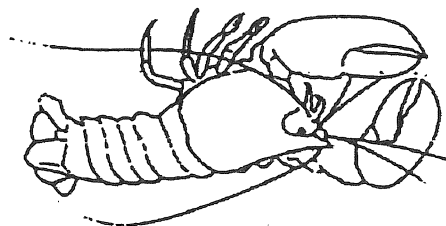


HUMAN BIOLOGICAL SCIENCE SERIES

Human Biological Science 1A

C. BOWDEN AND S. GILBERT

MARRON PUBLICATIONS



An introduction to Human Biological Science

37377/1

Human Biology is a very interesting subject that many students choose for a wide variety of reasons. Your answers to the following questions will be used to help your teacher deliver a more relevant and meaningful Human Biology course.

List some reasons why you selected Human Biology?

What would you like to do or find out about in Human Biology?

What are your future aspirations and how will studying Human Biology help you to achieve this?

What sports do you like to watch and/or play?

What hobbies and other interests do you have?

Do you have a part-time job and if so what are your responsibilities?

Introduction

Human biology is a science. It is the body of knowledge relating to humans, and is concerned with finding out more about them. This knowledge is continually increasing as new discoveries are made, and existing knowledge is often modified, or even rejected, as new evidence accumulates.

Some of the main fields of human biology that we will be covering are outlined below. Write descriptions for what you feel each of the fields may cover. Note – extra lines have been included for you to right down your teachers description. *Teachers see Teachers Guide*

Anatomy – _____

Biochemistry – _____

Cytology – _____

Embryology – _____

Genetics – _____

Nutrition – _____

Physiology – _____

Characteristics of living things

Biologists have found that living things have certain features in common. These features can be used to decide whether a thing is living or not. You may be familiar with the anagram MR GREEN and the seven characteristics of living things. *Teachers see Teachers Guide*

Movement — _____
 Reproduction — _____
 Growth — _____
 Response — _____
 Excretion — _____
 Energy — _____
 Nutrition — _____

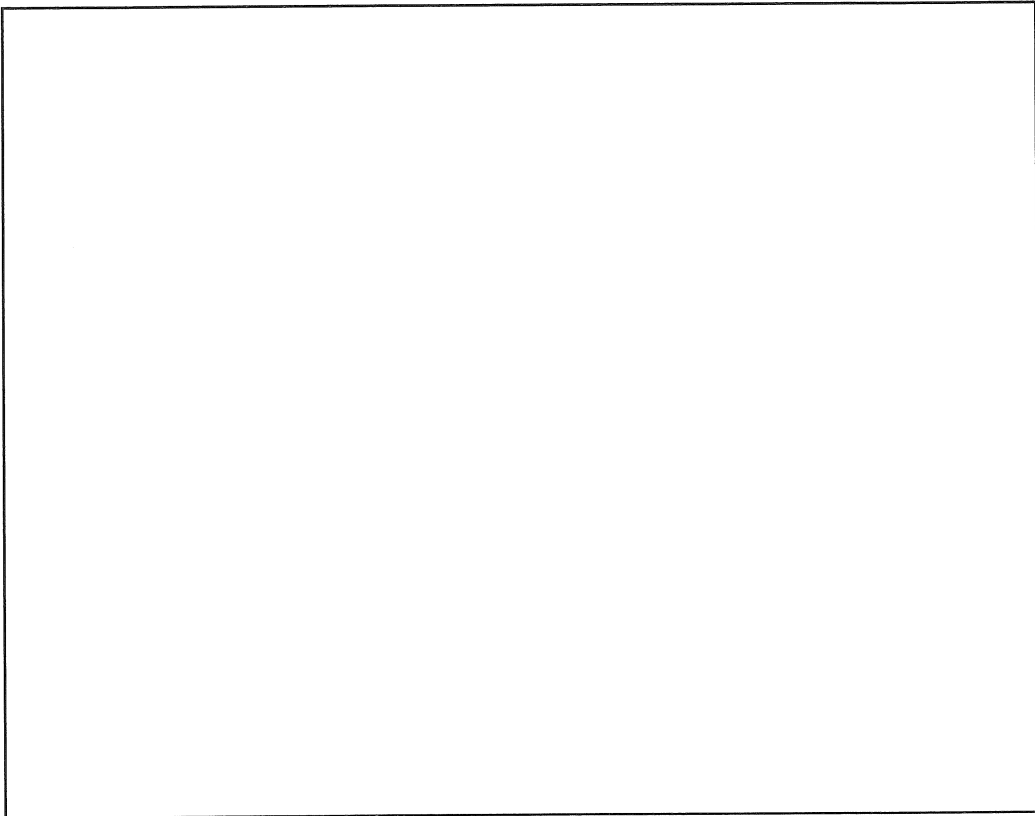
The characteristics mentioned above are generally taught in lower school. For this unit the life processes are outlined below. Write a short statement about each of these and create an anagram to remember them (you can reorder them to make up the anagram). *Teachers see Teachers Guide*

Metabolism — _____
 Ingestion — _____
 Excretion — _____
 Reproduction — _____
 Movement — _____
 Responsiveness — _____
 Growth — _____
 Differentiation — _____
 Anagram — _____

The simple cell

All living things are made up of a diverse range of different sized and shaped cells. Cells are the building blocks of life and virtually all living things are made up of cells. Generalised cells have a cell membrane, nucleus and cytoplasm. The cell membrane surrounds the cell and controls the movement of materials into and out of it. The nucleus controls the shape, size and functions of the cell, while the cytoplasm is where most of the cells chemical reactions take place.

In the space below draw a generalised diagram of a typical animal cell and label the nucleus, cytoplasm and cell membrane.

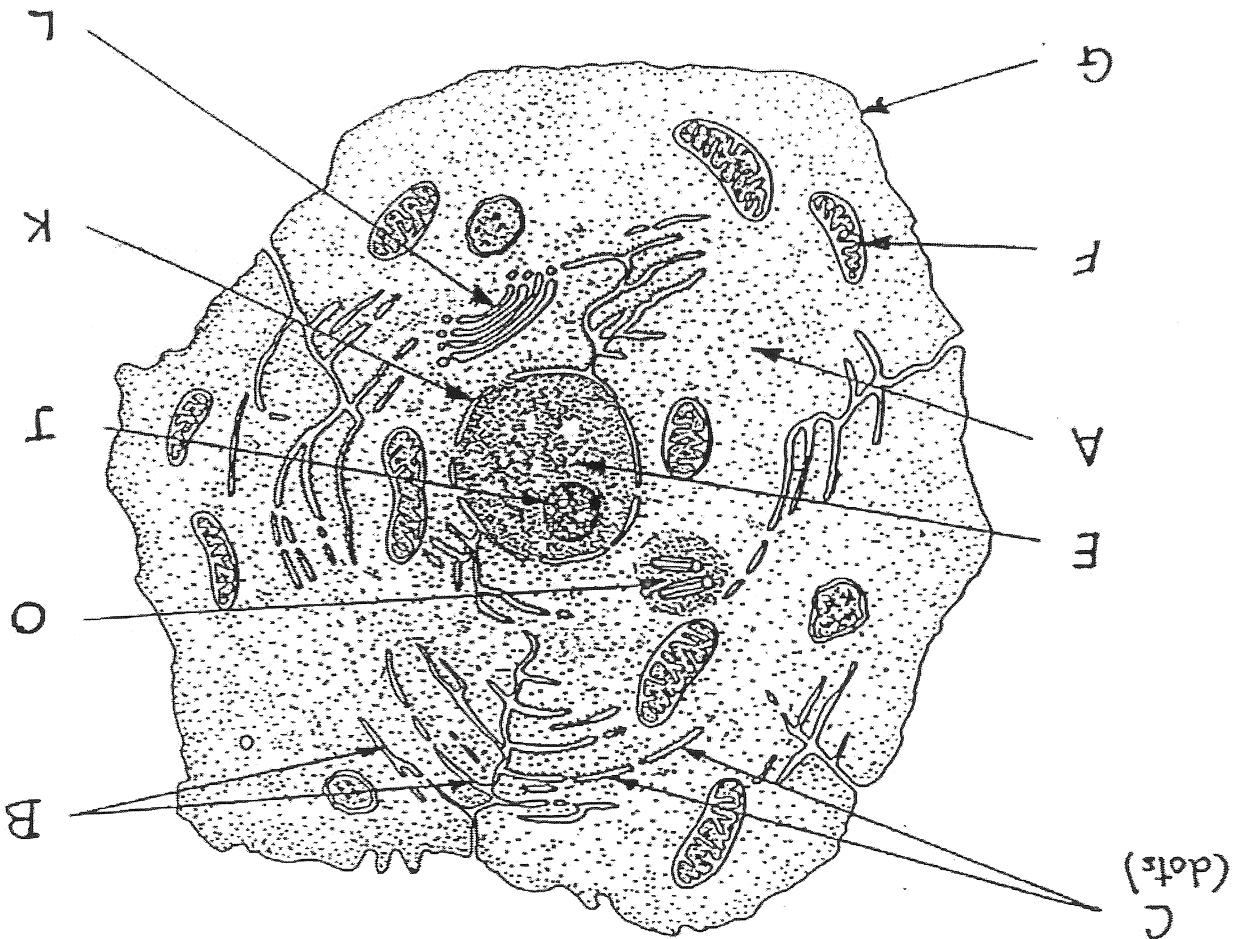


The complex cell

Cells are made up of various smaller building blocks called organelles. Organelles are responsible for carrying out particular 'functions' in the cell. Although varying in many ways all animal cells have a number of standard organelles.

With the aid of your teacher and other resources label the following structures on the diagram of a generalised human cell;

- Cell membrane.
- Cytoplasm.
- Endoplasmic reticulum.
- Nucleus.
- Nuclear membrane.
- Nucleolus.
- Golgi apparatus (body).
- Mitochondria.
- Ribosomes.
- Centrioles.



If you have access to the internet go to the URL www.biopic.co.uk/cellcity/ and take the tour of the cell.

A model cell

Your teacher may be able to show you a plastic model of a cell or diagrams of other generalised cells. You may notice that each of the organelles seem to have a certain shape and in many cases location within the cell.

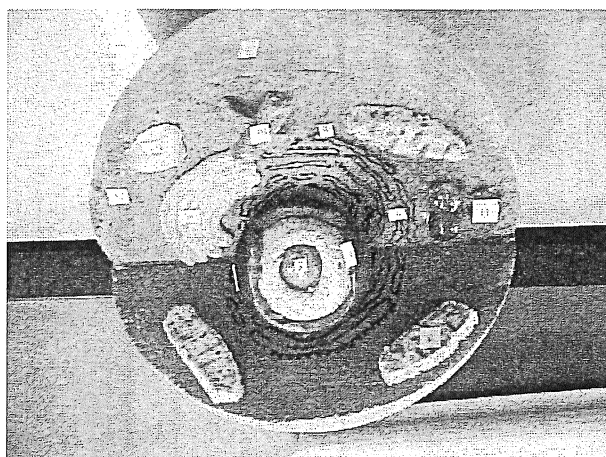
Your task is to design and make a model cell. Your cell should display the following labelled organelles. Labels can be either 'stickers' or 'flags' next to or attached to each organelle, or a separate labelled map/key.

- Cytoplasm
- Cell membrane
- Mitochondria
- Ribosomes
- Nucleus
- Endoplasmic reticulum
- Golgi apparatus
- Centrioles
- Lysosomes
- Cilia

As part of your planning you should discuss with your teacher the materials required and finished size of your model. If your model is likely to be biodegradable eg decompose/become mouldy you should arrange to produce it just before the day it is due.

Some suggestions for the main structure of the cell are, polystyrene floats, ice cream containers and cardboard. Suggestions for the organelles are plasticine, clay, dough, lollies eg smarties, liquorice, jelly beans, pasta, screws/nuts/bolts/wire, coloured and folded paper, buttons, straws, pipe cleaners etc.

Some pictures of some sample cell models.



On the due date the class will 'mark' each model using a marking system with the following criteria. *Teachers see Teachers Guide*

Microscopes

The human body contains billions of cells which are arranged into tissues. Since your cells are so small the naked eye can reveal very little about their structure and function. Compound light microscopes are very common and have a magnification of up to X1500. Samples can be quickly prepared and living cells can be viewed. However with many small structures the amount of resolution (detail visible) is limited. When greater magnification and resolution is required electron microscopes are used. The electron microscope can magnify up to X1 000 000, however they are very expensive and can only magnify dead specially prepared samples. The compound light microscope is used in schools to help understand cells, their structure and function.

Read and carefully follow the instructions below, before completing the activity because microscopes are delicate and expensive pieces of scientific equipment.

Parts of a microscope

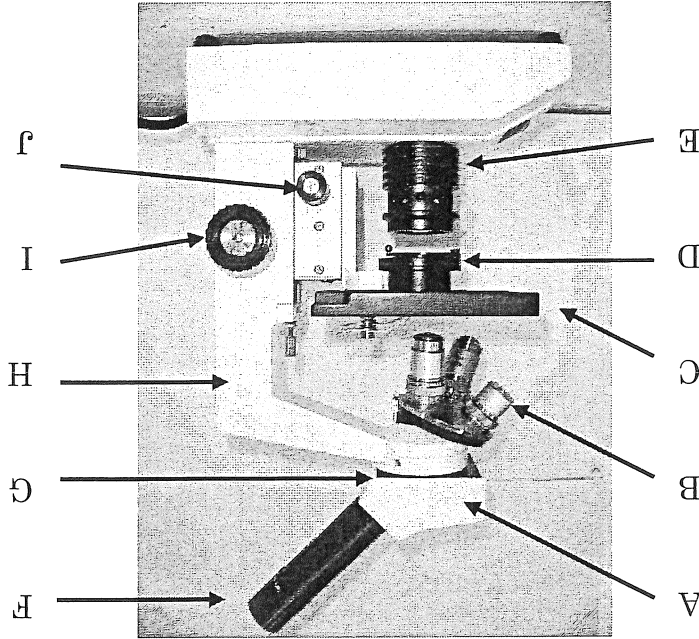
Aim – To use a microscope to focus on some prepared slides.

Materials

- Microscope
- Microscope lamp
- Prepared slides (Daphnia, Drosophila, Amoeba etc)

Procedure

- 1) Examine and identify the parts of a monocular compound microscope and then label the photo below. (Remember that microscopes will vary, but the basic components and functions will stay the same).



Focusing under low power

- a) Turn the microscope so that the _____ is facing me.
- b) Engage the _____ power objective (the shortest lens).
- c) Take out the _____ and stand it upright on the bench.
- d) Turn on the _____ if using one.
- e) Use the _____ to reflect light into the objective. Look down the body tube (barrel) to set the mirror correctly.
- f) Use the _____ to obtain a circle of light just as large as the field of view.
- g) Replace the ocular.
- h) Place the slide on the _____ and attach it with the _____.
- i) Using the _____, wind the objective down WHILE LOOKING AT THE STAGE FROM THE SIDE. Make sure the _____ will not hit the slide.
- j) Look into the ocular. Using the _____, wind the barrel up towards my eye (I may have to move the slide around a little to see if I am close to the focus). If I have wound the objective up more than one centimetre above the stage, I have gone too far, and should repeat steps (i) & (j).
- k) Use the _____ to obtain a clear image (once focused on low power do not move the microscope of adjusters – this will make it easier to focus under high power).

Focusing under high power

- l) Looking from the side, SLOWLY rotate the nosepiece/turret to engage the high power lens (the long one). Make sure the lens does not hit the slide.
 - m) Look into the ocular and use the fine focus ONLY to focus, only wind upwards never downwards.
 - n) If I have ANY problems I should see my teacher.
- Use this procedure to look at a number of prepared slides supplied by your teacher.

Prepared slides

It is important to be able to identify and draw biological diagrams of cells, tissues and specimens viewed under the microscope.

Aim- To view and draw biological diagrams of some prepared slides

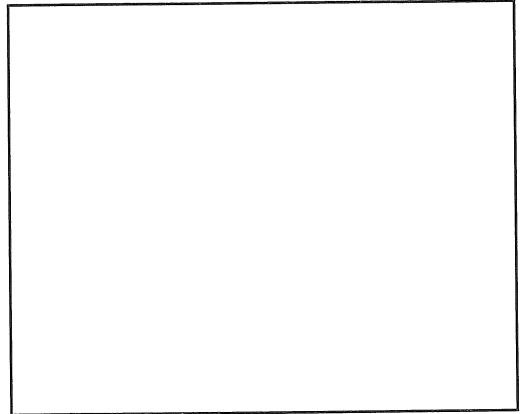
Materials

- Microscope.
- Microscope lamp.
- Prepared slides of tissues, blood, Amoeba, Euglena and Paramecium.

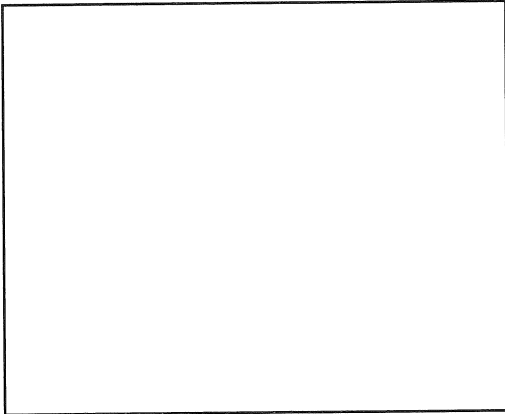
Procedure

- 1) Set up microscope and select a prepared slide.
- 2) Select a specimen of your choice and draw a biological drawing, your drawing should be;
 - In pencil and large enough to be easily viewable.
 - Show the magnification.
 - State the name of the specimen and label with straight lines.
 - Have a short written description.

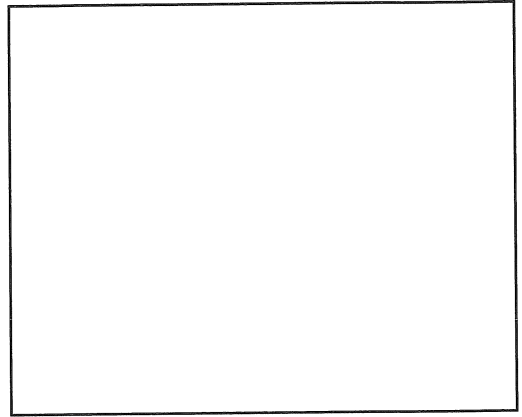
Amoeba – magnification x ____



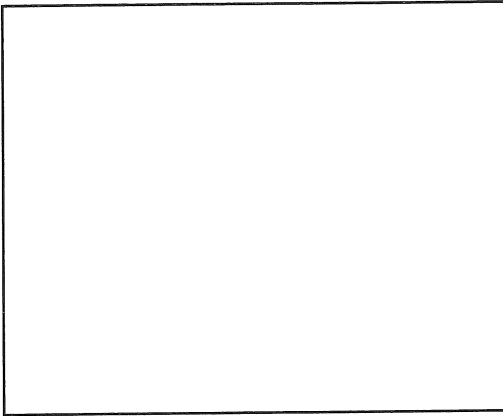
Euglena – magnification x ____



Blood – magnification x ____

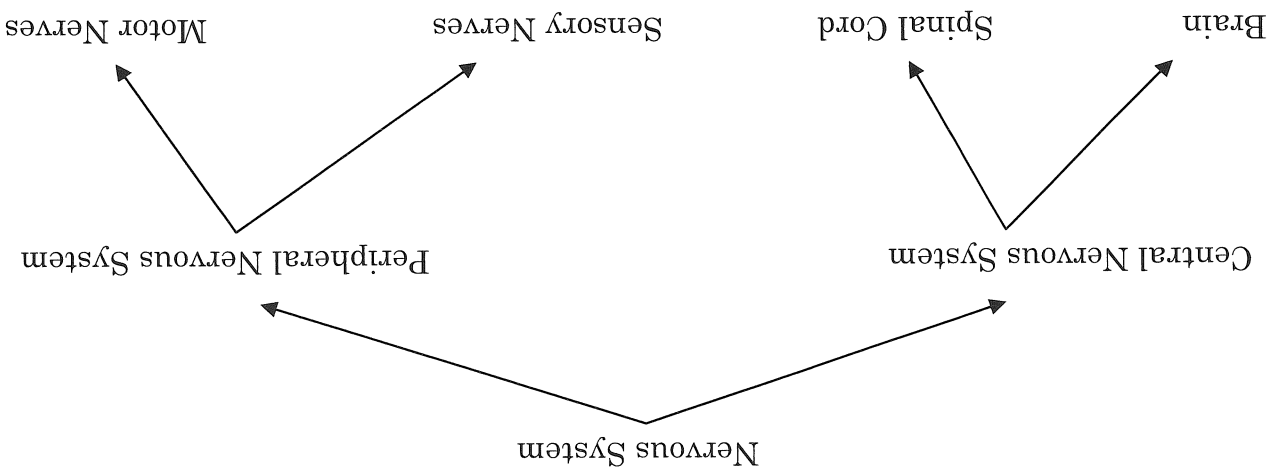


Paramecium– magnification x ____



The nervous system

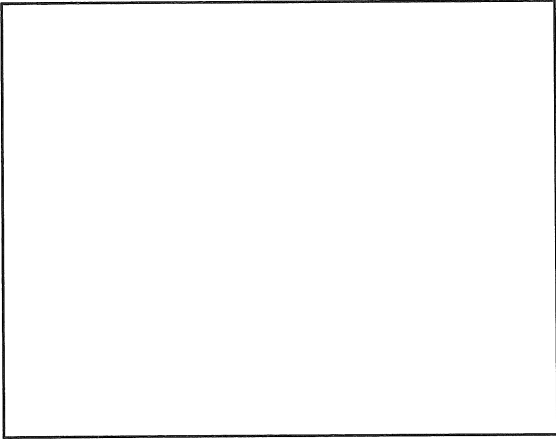
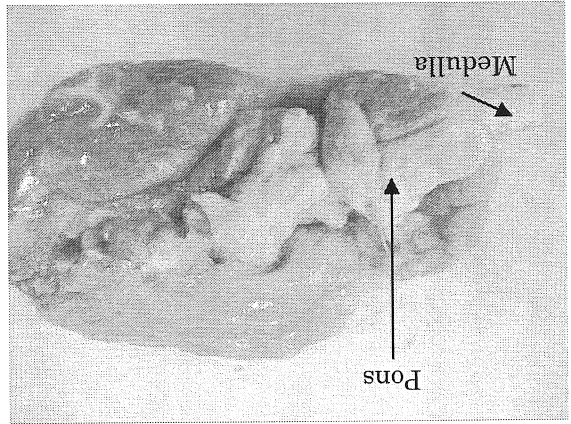
The following diagram gives a general overview of the nervous system.



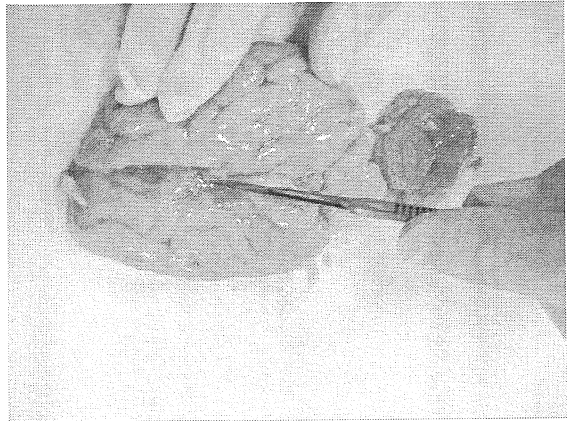
The central nervous system consists of the brain and spinal cord. The brain is one of the most complex organs in the human body and is made up of billions of neurons (nerve cells). The brain is responsible for processing information from the senses and initiating the appropriate response. Other functions of the brain include control of body functions like heart rate, breathing rate and body temperature. It is also responsible for complex thought like imagination, reasoning, emotions and memory.

One of the best ways to find out the structure and function of the brain is to complete a brain dissection.

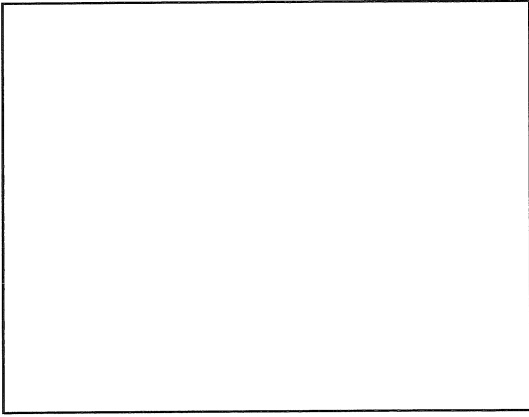
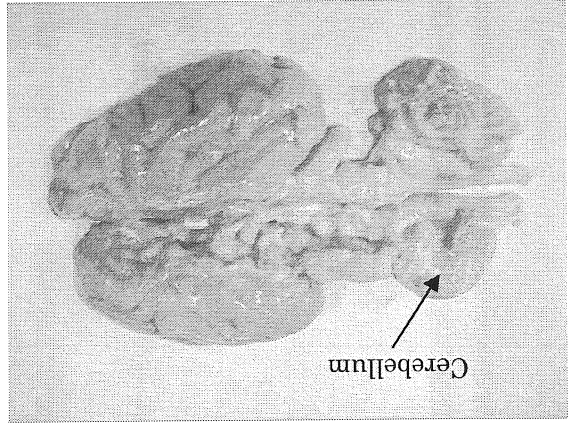
- 3) Turn the brain over and draw a sketch of the brain to show the parts in this and the previous diagram (pons and medulla).



- 4) Cut the brain longitudinally (down the middle) as shown in the photograph below.



- 5) You should be able to see the 'leaf-like' structure of the cerebellum at the rear of the brain. Draw it in the space provided.



Part of Brain	Function
Cerebral Cortex	
Cerebellum	
Spinal Cord	
Pons	
Medulla	

2. Humans have a larger area of cerebral cortex than a sheep. Why do you think this is so?

3. A blow to the back of the head can seriously effect your coordination. Taking into consideration the location of parts of the brain, what may cause this?

4. Why is the spinal cord at the base of the brain?

5. A number of people have died in fights by hitting their heads on the road or pavement. A severe blow to the head can cause the brain to swell, considering the location of the brain why may this cause severe brain damage or death? How can doctors try to overcome this problem?

6. What could you do to prevent yourself being involved in this type of situation?

7. What is the maximum jail term under the new "One punch" legislation?

8. The senses are an important part of the nervous system. They are some of the major receptors that provide the body with information about changes in the environment. The five major senses include

Detection of different stimuli

As mentioned in the notes the human body has a range of receptors, these receptors are not evenly distributed over the body. This means that particular areas of the body will be able to detect changes in the environment and others will not. An obvious example here is light falling in the eye will be detected (by photoreceptors), however light falling on the back of your neck will not be detected (there are no photoreceptors there). In this experiment we will look at other changes in the environment and the body's ability to detect them.

Aim – To test and record, the location and density, of taste and touch receptors on the body.

Materials

- Sterile cotton buds and blindfold
- Salt solution (15%)
- Sugar solution (15%)
- Quinine solution (0.01%) or Bitrex (available from chemists)
- 3 sterile Petri dishes
- Ruler and dividers

Taste Discrimination

Procedure and questions

1. Collect the sugar solution and place some in the bottom of a Petri dish. Using one of the cotton buds, dip it into the sugar solution and dab the cotton bud on the tip of the subjects tongue and at the back of the tongue. Have the subject record their taste sensations in these two positions (eg very sweet, sweet or no sensation).

2. Was there any difference in the sensations of the two areas of the tongue? If so explain why this may have occurred.

3. Repeat the above procedure using the salt and the quinine (bitter) solutions, making sure you use different cotton buds and that the subjects rinses their mouth between tests. Record your results in the space below. Did your tongue respond to the different taste sensations in a similar way? Explain your results in terms of receptors in the tongue.

4. On the photo above indicate the main areas where the different taste sensations were detected.

Questions

1. Which area of the skin had the most correct responses?

2. Which area of the skin had the least correct responses?

3. Which area of the skin would you expect to have the highest concentration of touch receptors explain?

4. Which areas of the skin would you expect to have the lowest concentration of touch receptors explain?

5. Is there a relationship between the number of receptors and the tasks carried out by the different areas of the skin?

6. Why were the one and two points arranged in a random order?

7. Why did we blindfold the subject?

8. Describe what you have learned about the distribution of touch and taste receptors over the body.

Functions of different parts of the eye

Your teacher will arrange for you to have access to a suitable text, or the library, so that you can complete the following table to describe the functions of the various parts of the eye.

	Structure	Function
Pupil		
Iris		
Cornea		
Ciliary muscles		
Retina		
Choroid coat		
Sclerotic coat		
Optic nerve		
Vitreous humour		
Aqueous humour		

8) Place each half of the eye with the cut side up on the dissecting tray.

9) You will notice a jelly like fluid in the halves, what is its name and function?

10) Looking at the rear part of the eye, locate the retina. You may notice an area where the blood vessels of the retina leave the eye. What is the name of this spot and what is it's disadvantage?

11) Look at the front part of the eye and see if you can locate the ciliary muscles and ligaments which "hold" the lens. What do they look like?

12) Remove the lens carefully and place it on a microscope slide. Place this microscope slide on some news print.

13) Describe the appearance of the newsprint through the lens.

14) Follow your teacher's directions and clean/return the equipment.

How good is your vision?

Aim - To test your vision using a standard eye chart.

Materials

- Standard eye chart (*Teachers see Teachers Guide*)
- 1 metre rule
- Subject

Procedure

- 1) Work in pairs, one will be the 'SUBJECT' the other the 'EXPERIMENTER'.
- 2) Measure out a distance of 20 feet (3.3 feet = 1 metre) on the classroom floor.
- 3) The experimenter should hold the eye chart 20 feet from the subject.
- 4) Each eye should be tested separately with your palm held over one eye while the other is tested.
- 5) Have the subject cover their left eye and read each line out aloud.
- 6) The experimenter should record the results in the table provided (Use the subject's workbook).
- 7) Repeat step (5) for the left eye but have the subject read each line backwards.
- 8) Swap roles and repeat steps (3) to (7).

Results

Line	Right eye number correct		Left eye number correct	
	Possible	Actual	Possible	Actual
Distance 50				
Distance 40				
Distance 30				
Distance 20				
Distance 15				
Distance 10				
Total 43				

Questions

1. Was there any variation between your left and right eyes? If so explain what could cause this?

2. A person with 'normal vision' should be able to read the distance 20 line at 20 feet without mistakes. Do you have 20/20 vision? Explain.

	Structure	Function
--	-----------	----------

Pinna		
Auditory Canal		
Ear drum		
Hammer, anvil, stirrup		
Cochlea		
Auditory Nerve		
Eustachian Tube		
Semi circular canals		

Question The sense of hearing is very important to humans. Explain using three examples why hearing is so important.

Reaction time activity one – alphabet race (part a)

Aim - To determine the time taken to touch a series of labelled squares in order.

Materials

- Fingers
- Stopwatch
- Workbook
- Partner
- Alphabet grids (*Teachers see Teachers Guide*)

Procedure -

- 1) You will need a partner; one will be the **TIMER**, the other the **SUBJECT**.
- 2) Use the practice grid to trial the procedure. Your teacher will give you the test grids when you are ready to proceed.
- 3) Cover up the alphabet grids. (Studying the series of grids before the activity will lead to false and misleading results)
- 4) Timer uncovers grid 1 and starts the stopwatch.
- 5) Subject uses forefinger to touch each of the letters in order from A to L as fast as possible.
- 6) Timer stops stopwatch when the subject touches the 'L' and records time in space provided below the grid in subject's workbook.
- 7) Repeat steps 3) to 6) for grids 2, 3 and 4.
- 8) Swap roles and repeat the activity.
- 9) Calculate your average time taken to complete a grid and your rating.
- 10) Record your average time and rating in the space provided.

Results

Grid	Time taken to complete (seconds)
1	
2	
3	
4	
Average	

average time (seconds)	rating	
less than 5	excellent	
5 - 7	good	
7 - 9	average	
more than 9	steady eddy	My rating =
		My time =

Reaction time activity one – alphabet race (part b)

Aim - To see how experience in a particular situation improves reaction time.

Materials

- Fingers
- Stopwatch
- Workbook
- Partner

Procedure -

- 1) Read the procedure in full before commencing.
- 2) Cover up alphabet race grid 4.
- 3) Timer uncovers grid 4 and starts the stopwatch.
- 4) Subject uses forefinger to touch each of the letters in order from A to L as quickly as possible.
- 5) Timer stops stopwatch when the subject touches the 'L' and records time in the table provided below.
- 6) Allow subject to practice grid four 10 times.
- 7) Repeat steps 3) to g) eg. Practice 10 times between each recorded trial.
- 8) Swap roles and repeat the activity.

Results

Trial	Time (seconds)
Initial	
Second	
Third	
Fourth	
Fifth	

Questions

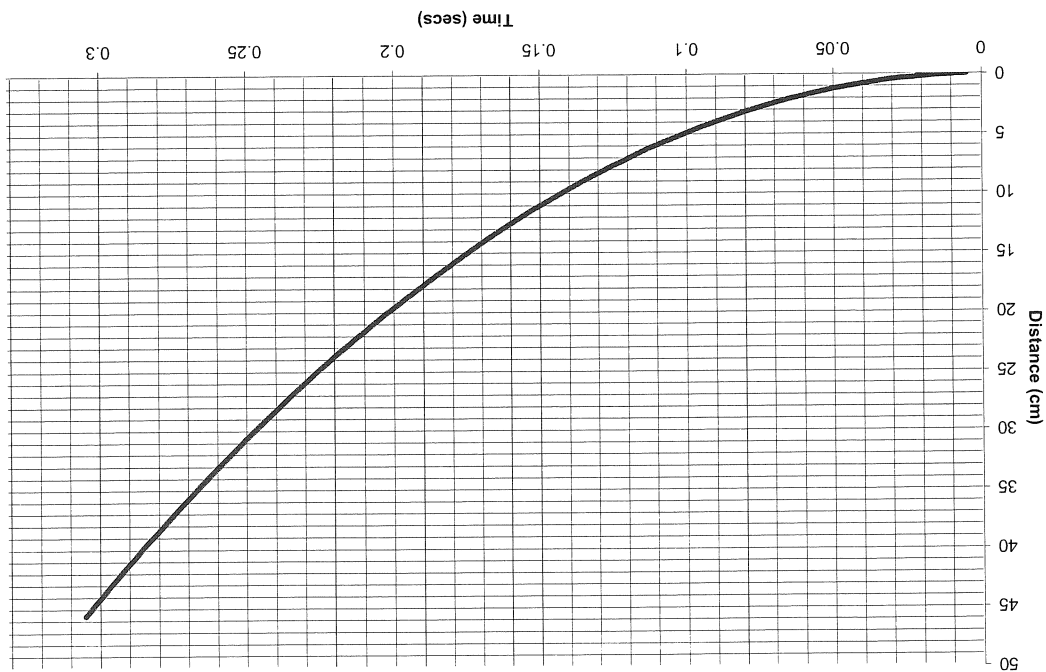
1. Describe and explain any trends in your results to part B.

2. Use an example where practice may reduce a reaction time and increase survival or quality of life.

Procedure - part 2

10) Use the graph and your average catching distance as calculated in part 1, to determine your reaction time and record the result in the space provided. Use a ruler and pencil to draw on the graph.

Distance vs Reaction Time Graph



My reaction time is _____

11) Your teacher will provide a table on the board for the class to enter their average reaction times. Once this has been completed, copy these results into the table below, and calculate the class average reaction time.

	Name	Reaction time		Name	Reaction time
1			14		
2			15		
3			16		
4			17		
5			18		
6			19		
7			20		
8			21		
9			22		
10			23		
11			24		
12			25		
13			Class average		

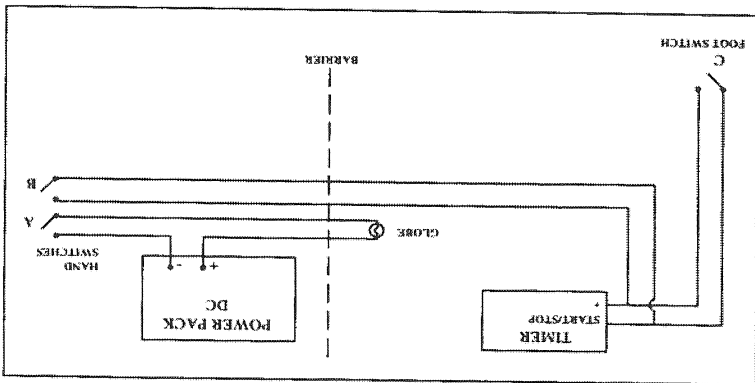
Reaction time activity three – braking time part a

Aim - To measure the reaction time of your braking foot.

Materials

- Power pack
- Connecting wires
- Electric timer
- 3 contact key switches
- Screen
- 12v light bulb & holder

Circuit diagram



Procedure

- 1) You will need a partner; one of you will be the SUBJECT, the other the EXPERIMENTER.
- 2) Set up the materials as shown in the circuit diagram above. (Your teacher may have a demonstration circuit set up for you to use or copy).
- 3) The subject should sit on a chair with their right foot flat on the ground next to the foot switch (brake pedal). The light bulb should be in view.
- 4) The experimenter should have their hand and the hand switch hidden from the view of the subject.
- 5) Without warning, the experimenter should quickly tap the hand switches A & B. This will cause the light bulb to flash and the timer to start.
- 6) The subject should quickly lift their foot from the ground and tap the foot switch C. This will stop the timer.
- 7) Record your reaction time in the table below, repeat the activity three more times and calculate your average reaction time. (You will have to reset the timer each time).
- 8) The subject and experimenter should swap roles and repeat (2) to (7).

Results

Trial	Reaction time (seconds)
1	
2	
3	
4	
Average	

- 7) Have the experimenter SMS/Text a friend a reasonably long message.
8) During this press the light switch and record the reaction time.
9) Repeat this twice and record in the results table.
10) Combine both the questions and SMS/Text, press the light switch and record the reaction time.
11) Repeat this twice and record in the results table.
12) The subject and experimenter should swap roles and repeat the activity.

Results

Reaction Time (Seconds)				Distraction	Questions	SMS	Questions and SMS
				Trial 1	Trial 2	Trial 3	Average

Questions

1. How does your reaction time for braking distance without distractions compare to your reaction time for braking with distractions?

2. Which type of distraction/s increased your reaction time the most?

3. Why do you think this type of distraction slowed down your reaction time?

4. Did you find that you had to concentrate harder when questions and SMS were introduced into the test? Explain.

4. Complete the following.

Bacteria

Description —

Examples of diseases caused by bacteria —

Viruses

Description —

Examples of diseases caused by viruses —

Fungi

Description —

Examples of diseases caused by fungi —

Animal Parasites

Description —

Examples of diseases caused by animal parasites —

You have seen a number of tables in this book already, but in this case you are to draw your

You have seen a number of tables in this book already, but in this case you are to draw your

You have seen a number of tables in this book already, but in this case you are to draw your

- You have seen a number of tables in this book already, but in this case you are to draw your

100

Bird flu

Your teacher will give you a copy of an article on bird flu (*Teachers see Teachers Guide*). Read the article and then use the information to answer the questions that follow. This is a reading and comprehension activity. Once you have finished reading you will be asked to return the information sheet before answering the questions, you are not allowed to reread the article.

Questions

1. What is a pandemic?

2. How many people did the "Spanish Flu" kill?

3. What is the correct name for 'bird flu'?

4. What concerns do scientists have with the H5N1 avian influenza?

5. When was the last case of avian influenza in Australia?

6. Do you think that another pandemic involving a virus is inevitable? Explain.

Defence against disease and foreign materials

The ability of the body to form a defence against disease is essential. The bodies defence system has two **tasks**;

- Prevent pathogenic micro-organisms and foreign materials from entering the internal environment of the body.
 - Killing or inactivating pathogenic micro-organisms and foreign materials that do gain entry.
- The bodies defence system has two main '**parts**';
- The first is non-specific and works to prevent all micro-organisms and foreign materials from entering the internal environment.
 - The second is specific as it 'tailor makes' actions to target specific pathogens that have entered the internal environment.

Non-specific defence (Inflammation)

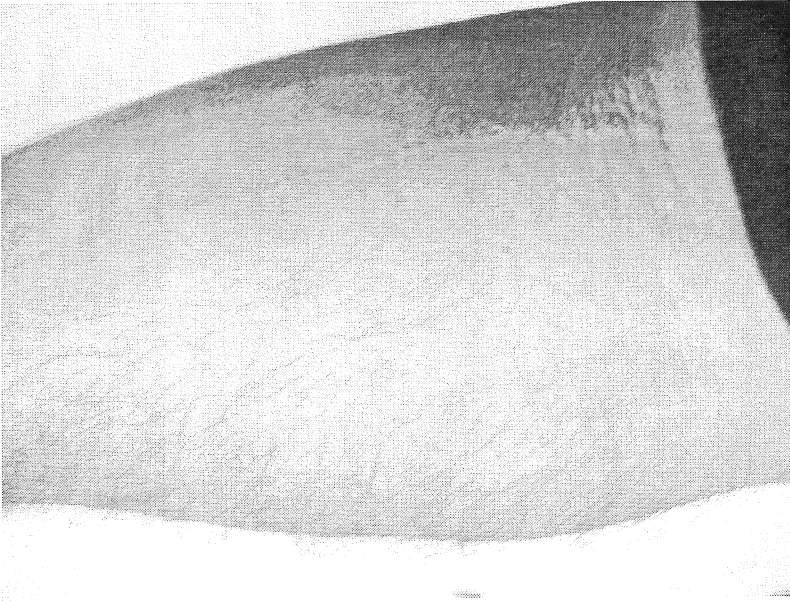
Activity

If you feel able, use your nails to scratch (without breaking the skin) the inside of your forearm. As you read on, observe your body's response and follow the explanation of what is occurring.

The human body has a complex series of actions that it can call on when foreign materials such as pathogens or non living materials eg dirt, splinters enter the body via the skin, mucous membranes or body openings eg nose. The first 'line of defence' once this occurs is called the **inflammation response**.

The inflammation response is a non-specific response that occurs whenever cells are damaged. Tissue damage can be caused by;

- Heat (sunburn).
- Chemicals (hay fever).
- Mechanical injury (scratches).
- Infection (pathogens).



Structural organisation of the human body

The Human Body begins from one cell formed at fertilisation, from this cell billions of cells are produced that work together to produce a functional organism. The size, shape and make-up of these cells differ according to the role that they perform.

All cells need to work together in a coordinated fashion to allow the human body to function efficiently. To do this the human body has arranged the cells into various levels of organisation, which includes individual cells, tissues, organs, systems and the organism.

Teachers see Teachers Guide

Cells –

T

Tissues –

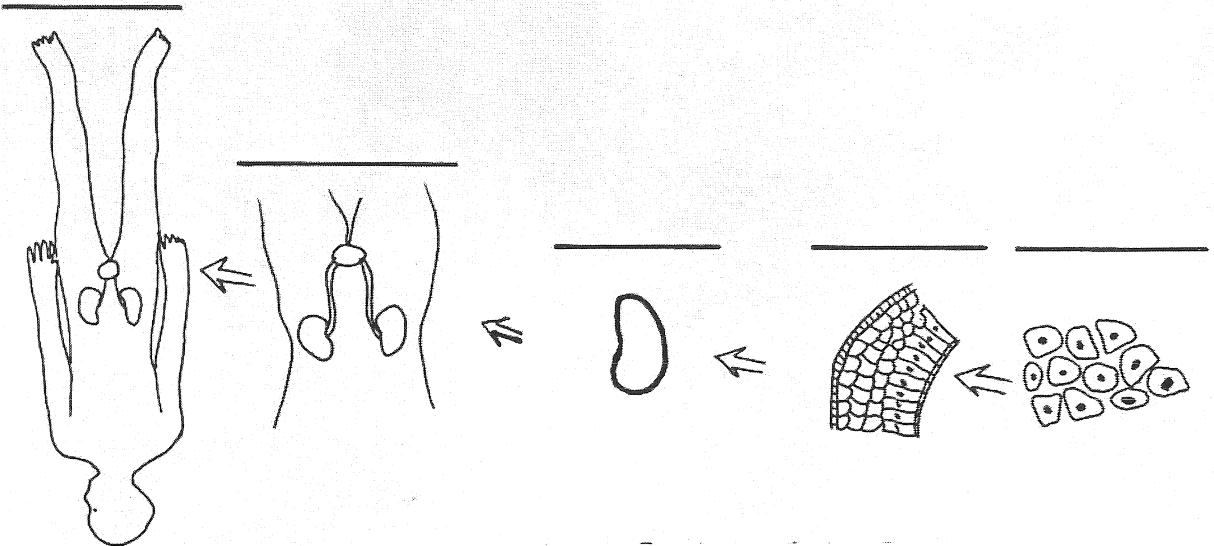
Organ –

System –

Organism

Summary

Label the diagram below with the terms described above.

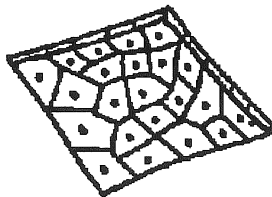
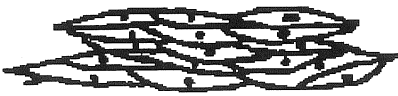
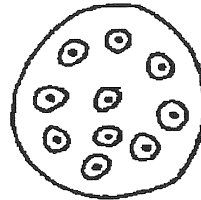


Nerve tissue is composed of specialized cells which not only receive stimuli but also conduct impulses to and from all parts of the body. Nervous cells work together to identify changes in the internal and external environments and then to coordinate the body's response to these changes. Some nerve cells can be over one metre in length to help speed up the rate of communication within the body. Nerve cells or neurons are long and string-like.

Connective tissues perform a variety of functions including support and protection. Connective tissue has the role of supporting and holding the body and other tissues together. Bone, cartilage, ligaments and tendons are all examples of connective tissue. Connective tissue is the most abundant and the most widely distributed of the tissues and includes ordinary loose connective tissue, fat tissue, dense fibrous tissue, cartilage, bone, blood, and lymph.

There are three types of **muscle tissue**: skeletal, smooth, and cardiac. Muscle tissue is the only tissue type that can contract. As a result muscle tissue is associated with movement, this could be the obvious movement of limbs or the less obvious movement of food or blood through the body. Skeletal muscle is a voluntary type of muscle tissue that is used in the contraction of skeletal parts. Smooth muscle is found in the walls of internal organs and blood vessels, it is an involuntary type. The cardiac muscle is found only in the walls of the heart and is involuntary in nature.

Identify and label each diagram as being Epithelial, Nervous, Connective or Muscle tissue.



Organs are the third level of organisation of the human body after cells and tissues. Different types of tissues join together in the body to form organs. Organs generally carry out a specific job using the different tissues. The heart for example has connective, muscular, nervous and epithelial tissues that help it to carry out its task of pumping blood around the body. Other organs include lungs, eyes, stomach etc all of which have a specific function. Organs do not work alone, they work with other organs in a coordinated way as part of a system. The circulatory system is composed of the heart, blood vessels and blood, which have the collective function of transporting materials around the body. Other organ systems

Questions

1 How are human cells different from each other?

2 Why do cells need to work together?

3 What are the levels of organisation of the human body?

4 What are the names of the generalised parts of the human cell?

5 What takes place in the cytoplasm of a human cell?

6 Define the term tissue and list the four basic tissue types.

Body systems

When looking at the structural organisation of the human body we have spent time studying the cell and how it functions. As mentioned earlier, cells make up tissues and tissues combine to produce organs. Some organs work together to produce systems and a large proportion of time in human biology is devoted to studying the bodies systems. The human body is made up of many systems and these systems operate together to enable the body to function efficiently. Some of the more well known body systems include the circulatory, muscular and skeletal, as the have easily identifiable structures. While others like the immune and endocrine are less obvious and their role may not be as clearly defined.

Below are listed the major systems of the body, complete the notes for each of these describing the main organs/tissues which make them up and the role that they play in the functioning of the body. *Teachers see Teachers Guide*

Circulatory

Muscular

Digestive

Respiratory

Skeletal

A system in detail – Nutrition and digestion

Nutrients

Humans are heterotrophs, that is, we eat other organisms to obtain food and energy for our own survival. Our body is a living machine that requires fuel, our type of fuel is food.

The food we eat contains nutrients, or molecules that provide the energy and material for:

- maintaining our existing cells
- building new cells
- repairing existing cells
- regulating cell & body processes
- maintaining a constant body temperature
- providing energy.

Every food type contains at least one of six kinds of compounds:

- carbohydrates
- proteins
- lipids (fats & oils)
- vitamins
- minerals
- water.

Four of these compounds; carbohydrates, proteins, fats and vitamins are organic compounds because they contain the elements carbon, hydrogen and oxygen. The two remaining compounds, minerals and water are inorganic compounds.

Use an information resource to complete the following table.

	Foods containing this compound	Nutrient(s) contained within the compound	Major function in the body
Carbohydrate			
Lipid			
Protein			
Vitamins & Minerals			
Water			

Fatty acids & glycerol

All lipids are composed of carbon, hydrogen and oxygen. Lipids are different to monosaccharides in that they have much less oxygen in relation to the carbon and hydrogen.

Lipids are derived from Fats (solid at room temperature) and oils (liquid at room temperature). A lipid is made of three fatty acids joined to a glycerol molecule. Fatty acids are chains of carbon and hydrogen compounds with a weak acid group attached to one end. Lipids that have only one fatty acid attached to the glycerol are called monoglycerides. Diglycerides are formed when there are two fatty acids attached and when there are three fatty acids it is called a triglyceride. 98% of lipids in foods are triglycerides.

Fatty acids are classified as either saturated or unsaturated. A saturated fatty acid has all of its carbon atoms joined (bonded) to hydrogen atoms. In an unsaturated fatty acid there are some 'spare' carbon atoms that could be used to bond more hydrogen atoms. A saturated fat mostly contains saturated fatty acids and is generally solid at room temperature. A polyunsaturated fat mostly contains polyunsaturated fatty acids and are generally liquids at room temperature. Fats provide twice as many kilojoules per gram as carbohydrates.

Vitamins

Vitamins are complex organic molecules that are needed by the body in very small amounts that serve as coenzymes (help enzymes in reactions).

Vitamins do not need to be digested, as they are small enough to be directly absorbed. With the single exception of vitamin D (made in the skin under direct sunlight), vitamins are not made by the body and must be obtained from food.

There are two main types of vitamins;

- Water-soluble - vitamins that cannot be stored in the body should be included in a balanced diet every day. Includes vitamins B and C.
- Fat-soluble - vitamins that can be stored in the fatty tissue of our bodies. Includes vitamins A, D, E, and K.

Like other essential nutrients, most vitamins can be obtained naturally by eating a balanced diet that includes fresh fruits, vegetables and meats. When the body does not receive a sufficient supply of vitamins, it can develop vitamin deficiency diseases, for example, scurvy once common among sailors, was caused by a lack of vitamin C in the diet.

Types of nutrients required by cells – summary table

Elements in nutrient	Source of nutrient	Types	Function(s)	Example(s)
Mono – saccharides				
Amino acids				
Fatty acids & glycerol				
Vitamins				
Minerals				
Water				

Testing foods for nutrients (Part 1)

Aim – To complete a series of food tests on known food solutions.

Materials

- Solutions of glucose, starch, protein
- Fat eg butter / lard
- Plastic pipettes
- White depression tile + marking pen
- Test tube rack + 4 test tubes + test tube holder + test tube cleaning brush
- Brown paper
- Testape
- Potassium Iodide solution (dropper bottle)
- Benedict's solution (dropper bottle)
- Biurets Reagent 1 - 'Strong Alkali Solution' - (1M sodium carbonate solution)
- Reagent 2 - 'Dilute Copper Sulfate Solution' (0.1M copper-II-sulfate solution)

Procedure

- 1) Use the procedure from the testing foods for nutrient table. Because you are using pre-prepared solutions you do not need to add water to the samples.
- 2) For each solution make sure that you can identify the 'positive' colour change that indicates that particular nutrient.

Results

Tick each of the boxes in the table below if you can identify the 'positive' colour change that indicates that particular nutrient.
Your teacher may ask you to complete one of the tests in front of them to demonstrate your knowledge of the test and the positive result.

Nutrient	Test completed	+ve colour change observed = ✓
Glucose		
Starch		
Protein		
Lipid		

Nutrient deficiencies

A deficiency in a particular nutrient can lead to a deficiency disease or particular symptoms.

Your teacher will allocate one or more of the minerals or vitamins in the table below to each of the members of the class. You are to research good food sources of this nutrient, its role in the body and the effects of a deficiency of this nutrient on the human body.

You will then give a short oral presentation to the class or use a 'jigsaw' or 'round table' to share the information with other members of the class.

You will need to be concise with you information and presentation.

Vitamin	Good food sources	Role in the body	Effect of a deficiency
A			
B1 (thiamine)			
B2 (riboflavin)			
B3 (niacin)			
B5 (pantothenic acid)			
B6 (pyridoxine)			
B9 (folate)			
B12			
C			
D			
E			
K			

Fast food internet activity

Imagine that your parents are taking you to a fast food / hamburger restaurant for tea/take away and they tell you to order what you like.

Step 1

Write down below what you would order. (If you don't eat these types of food your teacher will give you an 'average' meal to complete.)

Main meal (Burger(s)) - _____

Side order (Fries/onion rings) - _____

Drink (Cool drink/shake) - _____

Desert (Sundae/thick shake) - _____

Other - _____

Step 2

Go to McDonalds Website (www.mcdonalds.com.au) or Hungry Jacks website (www.hungryjacks.com.au) and complete the following table.

Step 3

Food item	Amount of energy (Kj)	Protein	Total fat	Total carbohydrate

Step 4

The recommended energy intake for a 15/16 year old is between 11 000 – 12 000 Kj per day. (This may vary depending on activity levels). Work out the percentage of the recommended energy intake that the meal you outlined above represents. (Show you working)

Overweight & obesity

Over the last two decades the number of Australians who are overweight or obese has almost doubled, Australia is now ranked as one of the fattest developed nations, closely following USA rates. It is estimated that 67% of adult men and 52% of women are overweight or obese. Males are more likely than females to be overweight, with almost half (48%) of adult males estimated to be overweight compared to 30% of females. The levels of obesity are higher in females, with 22% of females estimated to be obese compared to 19% of males. Besides genetic factors, overweight and obesity are caused by an energy imbalance; this is where over a considerable period of time energy intake (food) exceeds energy use (activity).

Use the grid on the following page to draw a line graph of the data below.

Note –

- On the vertical axis use a scale that measures from 0 to 65%.
- On the horizontal axis use a scale from 1980 to 2000.
- Use eight colours or similar to illustrate each line eg Underweight males, Obese females and provide a key.

Year	Body mass index		%Males	%Females
1980	Underweight	1.1	5.8	
	Healthy weight	51.5	67.0	
	Overweight but not obese	37.9	19.3	
	Obese	9.4	7.9	
1983	Underweight	1.2	3.8	
	Healthy weight	49.5	64.0	
	Overweight but not obese	40.4	21.8	
	Obese	8.9	10.4	
1989	Underweight	0.6	3.6	
	Healthy weight	46.2	59.5	
	Overweight but not obese	42.8	24.4	
	Obese	10.4	12.5	
1995	Underweight	0.7	2.3	
	Healthy weight	34.7	48.2	
	Overweight but not obese	45.7	30.3	
	Obese	18.9	19.1	
2000	Underweight	0.6	1.6	
	Healthy weight	32.1	46.5	
	Overweight but not obese	48.2	29.8	
	Obese	19.1	22.2	

Data source – <http://www.aihw.gov.au/dataonline/riskfactors/RF-Body-weight.xls#Print> - Summary Data!A1

How overweight and obesity is defined

Overweight and obesity is measured by using the Body Mass Index (BMI). This is calculated by dividing your weight in kilograms by your height in metres squared. An example of a BMI measurement is detailed below

If you weighed 80 kg and were 2 m tall, this is how you would calculate your BMI;

- Determine your height in metres squared ($2\text{ m} \times 2\text{ m} = 4\text{ m}^2$).
- Divide your weight (80 kg) by your height in metres squared (4 m^2).
- Your BMI would be 20.

BMI is an approximate measure of the best weight for health only. Overweight is measured at a BMI of 25 or more with obesity determined at a BMI of 30 or more.

Note – Body shape and ethnicity can have an effect on BMI, any concerns should be discussed with a nurse/doctor.

The first table below outlines the classification of different Adult BMI's. The second gives a more accurate comparison for adolescents at the cut offs for overweight and obese categories. (*Teachers see Teachers Guide*)

Table 1 – BMI classification

BMI Classification	<18.50
Underweight	18.50 - 24.99
Normal range	>25.00
Overweight	25.00 - 29.99
Pre-obese	30.00 - 34.99
Obese class 1	35.00 - 39.99
Obese class 2	>40.00
Obese class 3	

Table 2 - Classification of overweight and obesity adolescents

	23.29	23.94	28.30	29.11
	23.60	24.17	28.60	29.29
	23.90	24.37	28.88	29.43
	24.19	24.54	29.14	29.56
	24.46	24.70	29.41	26.69
	24.73	24.85	29.70	29.84
	25.00	25.00	30.00	30.00

McSlush & Slopper – An investigation of fat content

Aim – To determine the percentage of fat in a fast food hamburger and fries meal.

Materials

Per class (5 groups)

- 1 hamburger with cheese, large fries & large cola
- Blender, electronic scales & access to a refrigerator

Per group

- Bunsen burner, tripod, gauze mat and matches
- 500 ml beaker, 250 ml beaker, 100 ml beaker
- 200 ml graduated cylinder & stirring spoon
- 100 ml of water

Procedure

- 1) Weigh the burger and fries and record the weight in the results table.
- 2) Break up the burger and fries into small pieces and blend with the coke.
- 3) Evenly divide the mixture between each group by pouring part of the blended meal into a series of 500 ml beakers.

- 4) Add 100 ml of water to your portion of the blended meal and stir well.

- 5) Boil the meal mixture gently for 15 minutes.

- 6) Decant (pour) or strain (not filter) the liquid from the boiled mixture into the 250 ml beaker.

- 7) Allow the mixture to cool in the refrigerator until the fat solidifies at the top of the liquid. (It may need to be left overnight)

- 8) Weigh the 100 ml beaker.

- 9) Remove the beaker from the refrigerator and use the spoon to scoop the fat into the pre weighed 100 ml beaker.

- 10) Reweigh the beaker and calculate the weight of the fat.

- 11) Record the results in the table and use the class results to determine the weight and percentage of fat in the meal.

Results

Weight of burger & fries = _____

Group	1	2	3	4	5
Weight of 100 ml beaker					
Weight of 100 ml beaker + Fat					
Weight of fat					

Total weight of fat = _____

Percentage of fat in the burger & fries meal = _____

The basic digestive tube

The digestive system is comprised of specialised organs and tissues that are responsible for digestion and absorption of nutrients.

The sequence of events of digestion can be divided into four main processes; Ingestion, Digestion, Absorption and Egestion.

Each process has specialised organs associated with its processes.

Ingestion	Digestion	Absorption	Egestion
Mouth	Mouth	Small Intestine	Rectum
Tongue	Stomach	Large Intestine	Anus
Teeth	Small intestine	Liver	
	Salivary glands		
	Pancreas		
	Gall Bladder		

How long is the human digestive system?

Aim – To make a simple model to illustrate the length of the human digestive system

Materials

- 6 different coloured 'balls' of string/twine
- Scissors

Procedure

- 1) Use the scissors to cut the coloured string into pieces that can be tied together according to the measurements below. Remember that you need to allow some extra for the 'knots'.
- 2) Tie the pieces together and lay it out in a straight line.
- 3) Complete the table with a key for the colours that you used.
- 4) Write down the function of each part.

Results

Part	Length (cm)	Colour	Function of the part
Mouth	10		
Oesophagus	25		
Stomach	20		
Small Intestine	700		
Large Intestine	150		
Rectum	10		
Total		-----	

Questions

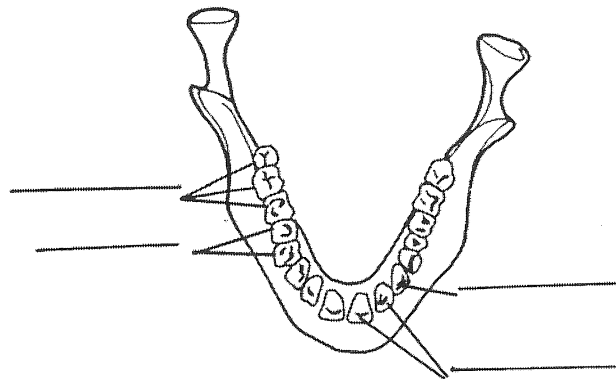
1. Why do you think different parts of the digestive tube are different lengths?
2. Why do you think the digestive tube needs to be so long?

Functions of the components of the digestive system

Component	Function
Salivary glands	
Mouth	
Oesophagus	
Stomach	
Pancreas	
Liver	
Gall bladder	
Small intestine	
Appendix	
Large intestine	
Rectum	
Anus	

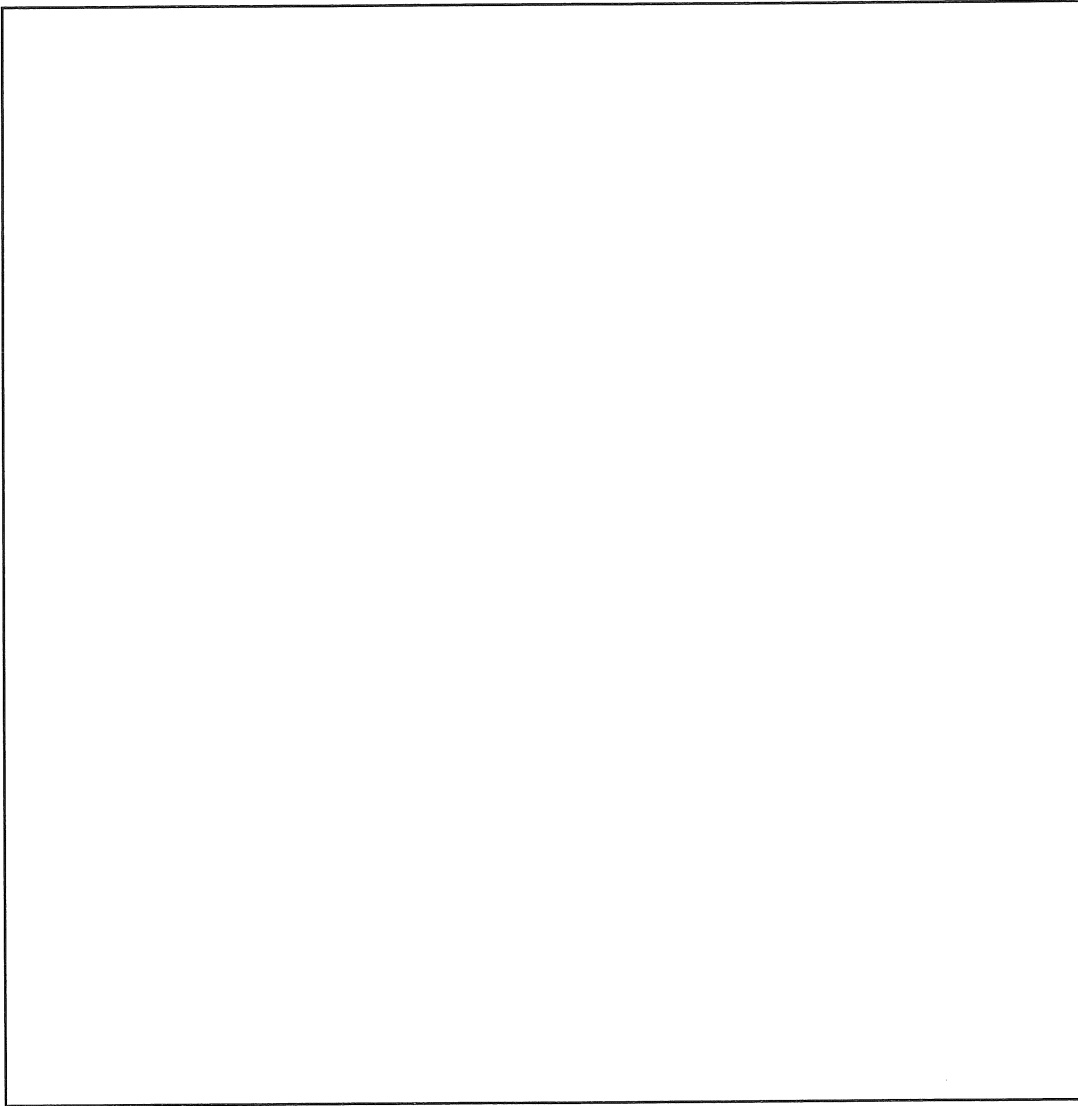
Location

Label each type of tooth on the diagram below.



Structure

Draw a diagram of a molar tooth and label the following
Crown, root, neck, enamel, dentine, pulp cavity/pulp, gum, bone, cementum, blood
vessels/nerves, root canal.



Chemical digestion

- Chemical digestion is the breakdown of small particles of food into small molecules that can be absorbed through cell membranes.
 - It occurs in the mouth, stomach and small intestine.
 - Digestion is controlled by enzymes.
 - The food or substances that enzymes act on are called substrates.
 - The new material that is made is called an end product
 - In general enzymes are named after their source and the substrate they act on. The named is developed by adding 'ase' as a suffix to the name of the substrate e.g. Pancreatic protease is an enzyme from the pancreas that breakdown proteins.
- The table below outlines the major digestive enzymes. Your teacher will discuss with you the degree of understanding and memorisation that is required. *Teachers see Teachers Guide Notes*

Enzyme	Source	Substrate	End product
Salivary amylase			
Gastric protease			
Pancreatic amylase			
Pancreatic protease			
Pancreatic lipase			
Peptidase			
Intestinal lipase			
Disaccharide enzymes			

Digestive Juice	Source	Function
Gastric HCl		
Bile salts		
Sodium Hydrogen Carbonate		

Procedure – Lesson 2

- 1) Remove the cling wrap and use your spoon to check the consistency of the jelly. If permitted by your teacher you can eat the jelly and pineapple.
- 2) Record your observations.

Observations

Explanation

The substance that causes jelly to thicken is a protein called 'Gelatin'. The bromelain in the fresh pineapple is a protease; this 'breaks down' the gelatine in the jelly stopping it from setting. The tinned pineapple was cooked during the canning process. The heat denatured the Bromelain, which stops it from working as a protease; therefore the gelatine was able to set.

Turning starches to sugars

Aim – To digest a starch containing food in the mouth to produce sugar. Saliva contains amylase, an enzyme that breaks down starch into sugar molecules.

Materials

- A low salt saline cracker or biscuit that contains no sugars.

Procedure:

- 1) Chew on the biscuit for 1 to 2 minutes.
- 2) Don't swallow the biscuit.
- 3) Swallow or dispose of the biscuit mush after 2 minutes.
- 4) Record your observations about the taste of the biscuit.

Observations

Explanation

At first the biscuit / bread would have tasted very bland or plain. However with continued chewing, the taste of the biscuit / bread should turn mildly sweet. This sweet taste is the result of the amylase converting the starch into sugars.

Absorption from the small intestine

Following the process of digestion there is a high concentration of nutrients in the lumen. To enable their absorption the small intestine has a number of features.

Teachers see Teachers Guide

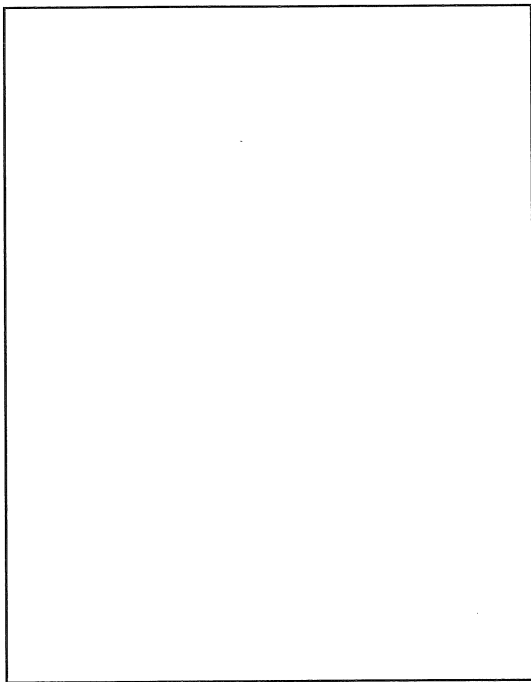
Notes

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-
-
-

Villi and absorption

Draw a labelled diagram of a villus highlighting each of the following;

- Epithelial absorptive cell.
- Epithelial secretory cell.
- Microvilli.
- Capillary network.
- Arteriole.
- Venule.
- Lacteal.
- Lymph duct.



Explain how the structures that comprise a villus individually and as a whole, aid in the small intestines role of absorption.

Absorption in the Large intestine

By the time the digested food/chyme reaches the large intestine the majority of nutrients have been absorbed. The large intestine is primarily responsible for absorbing water. Water-soluble vitamins and minerals are absorbed with the water.

Mucus secreting cells make mucus, which helps 'movement' as the water is removed. It also 'houses' bacteria that breakdown the remaining organic nutrients that were not absorbed, their 'waste product' is a gas either methane and/or hydrogen sulphide.

Some bacteria produce vitamins as a 'waste' product of their metabolism; these are absorbed by the intestinal cells.

Elimination / defecation

Teachers see Teachers Guide

Notes

-
-
-
-
-
-
-

Dietary disorders

Condition / disorder	Description	Cause	Part involved / affected	Treatment / cure
Vomiting				
Indigestion				
Gastric reflux				
Constipation				
Diarrhoea				
Peptic Ulcers				
Gastric Ulcers				
Irritable Bowel Syndrome				
Crones Disease				
Appendicitis				
Flatulence & belching				

12. Ans – Aerophagy

12. Qu –

13. Ans – Firmly clenching something like a pencil between the teeth

13. Qu –

14. Ans – Borborygmi

14. Qu –

15. Ans – Hunger, or by anxiety, or a fright: they are very common in irritable bowel

syndrome

15. Qu –

16. Ans – Times per day (ranging between 3 and 40 times)

16. Qu –

17. Ans – High fibre diet

17. Qu –

18. Ans – Normal bacterial fermentation of food residues in the colon

18. Qu –

19. Ans – Oligosaccharides

19. Qu –

20. Ans – Cabbage, brussel sprouts, cauliflower, turnips, onions, garlic, leeks and some

seeds such as fennel, sunflower and poppy

20. Qu –

21. Ans – Activated charcoal

21. Qu –

22. Ans – Lactose

22. Qu –

23. Ans – Narrow anus

23. Qu –

24. Ans – Indoles, skatoles and hydrogen sulphide

24. Qu –

25. Ans – Asafoetida

25. Qu –

26. Ans – Sulfur

26. Qu –

27. Ans – Chewing gum, smoking, or wearing loose dentures

27. Qu –

The human lifecycle

The human lifecycle on average in Australia currently takes on average 79 years for males and 84 years for females. With advances in medical technology this is increasing.

The human lifecycle has definite start (conception) and end (death) points, the period in between these can be divided into seven periods; embryo/foetus, baby/infant, toddler, child, adolescent/teenager, adult and geriatric (old age).

Conception

The lifecycle commences with the production of a new individual entity.

Sexual reproduction is a cellular event, it involves;

- The production of specialised reproductive cells called gametes.
- The placement of these gametes next to each other (coitus).
- The joining of the gametes (fertilisation) to form a zygote.
- The implantation of the zygote into the lining of the uterus.

The period of development inside the uterus is termed gestation.

In sexual reproduction gametes are made by male and females, these gametes join to form a new individual. The gametes are involved in transferring genetic information from parents to offspring and their production and union result in a number of genetic similarities and differences (variation) to each of the parents.

Fertilisation is the joining of gametes to form a zygote. Fertilisation can be external where the gametes join in the external environment. Internal fertilisation occurs inside the female body. Internal fertilisation is an advantage as it protects the gametes and zygote from drying out and interference by other external factors.

After 3 weeks of growth by mitosis the zygote is termed an embryo.

Embryo/foetus

Between the fourth and seventh week the embryo continues to increase in size as the organ systems develop. By the end of the seventh week it is approximately 3 cm long, noticeably human like in form and is now termed a foetus.

From the start of the third month (ninth week) to the end of the ninth month the foetus continues to increase in size and complexity. At the end of the ninth month, just before birth the foetus weighs on average 3.3 kg and is around 50 cm in length.

Baby/infant

The birth of a foetus is termed parturition and is preceded by a sequence of events called labour where the body prepares for and then expels the foetus, which is then termed a baby.

The baby/infant is totally reliant on its mother for protection, warmth, fluid/food and hygiene. Besides inbuilt reflex actions such as sucking, crying, swallowing and blinking it needs to learn all other skills required for survival.

Adolescence

From the moment of birth we continually grow until we reach physical maturity and adulthood. From birth to the age of 2 we grow at a very fast rate, for example at the age of 1 we are growing at an average rate of 16cm a year. (If we continued at that rate how tall would you be at the age of 20?). From 3 to 10 years our growth rate slows down and we are growing on average 5cm per year.

At around 11 years of age the growth patterns of males and females starts to differ. Females start a 2 year growth spurt some where between the ages of 11 and 14. During these two years their height will increase on average 15cm and their weight will increase by an average of 15 kg. The majority of girls go through this growth spurt during year 7 and 8 and are noticeably larger then their male counterparts at his time.

Males go through a similar 2 year growth spurt to females, however theirs occurs between the ages of 13 to 16. During these two years their height will increase on average 20 cm and their weight will increase by an average of 16 kg. This explains why in year 9 and 10 the boys become larger then the girls.

These changes in weight and height are an integral part of the period of transition from child to adult which is termed adolescence.

All systems of the body are involved in growth and change during adolescence. In addition to the obvious increase in size of the skeletal and muscular system, the circulatory changes as the heart grows to almost twice its childhood size and the capacity of the lungs increase markedly.

One of the systems which show the most obvious changes is the reproductive system. The reproductive system is underdeveloped during childhood primarily as it is not required.

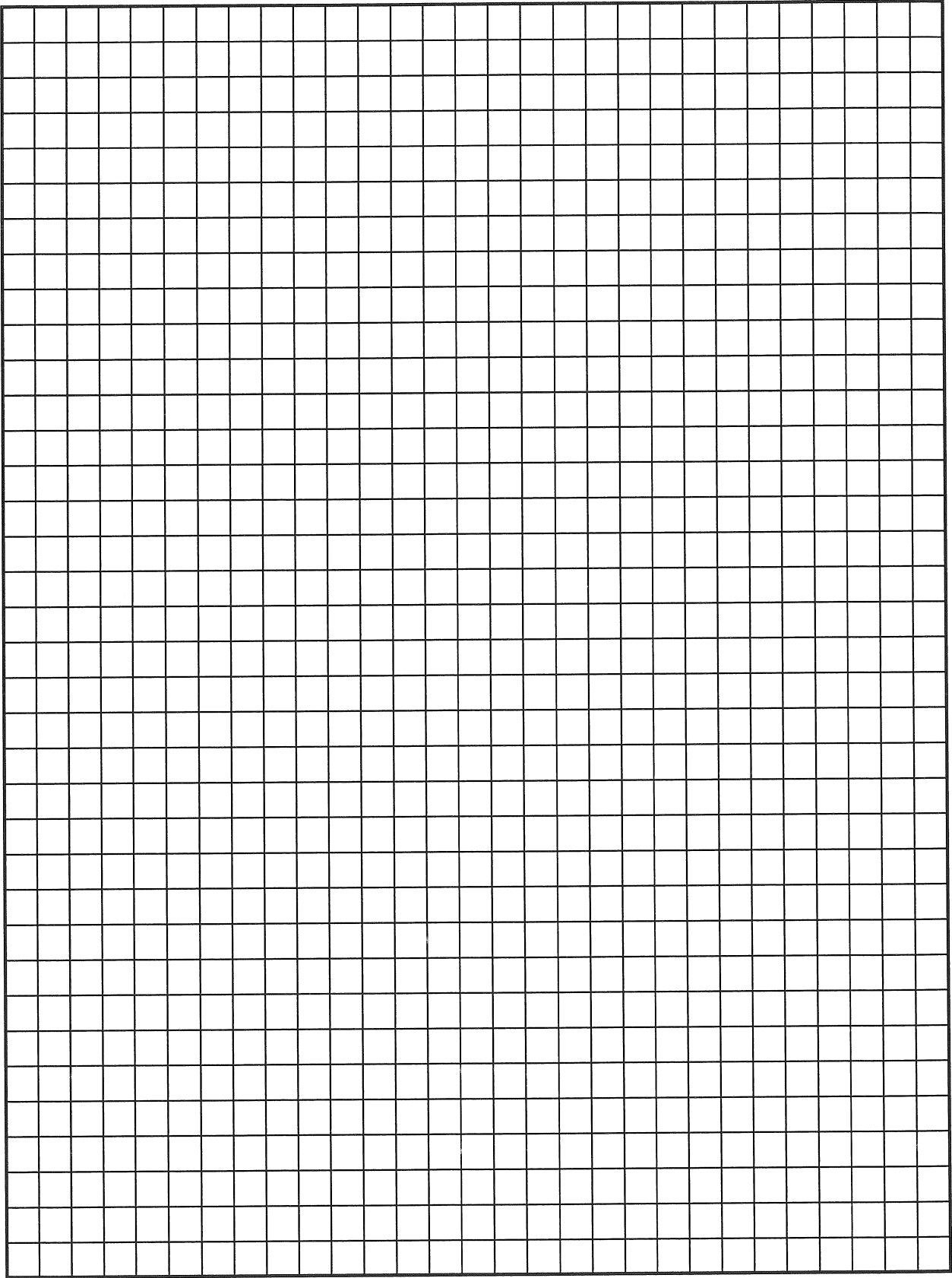
One of the phases of adolescence, called puberty, is when the reproductive system develops so that it is capable of making gametes. Males also require the ability of transferring them to the female and females the capability of carrying a baby and feeding it following birth.

Over the average of three years of puberty in male there is an increase in the size of the testes and scrotum, lengthening and thickening of the penis, ability to ejaculate semen, development of pubic, facial and body hair and changes in voice tone due to the enlarging of the larynx.

In females there is an increase and size and changes to the structure of the ovaries, uterus and vagina, and menarche (first menstrual cycle) occurs. In addition a number of secondary sexual characteristics develop;

- Development of the breasts.
- Development of pubic and some body hair
- Increase in body fat distribution to the hips and breasts.

Graph

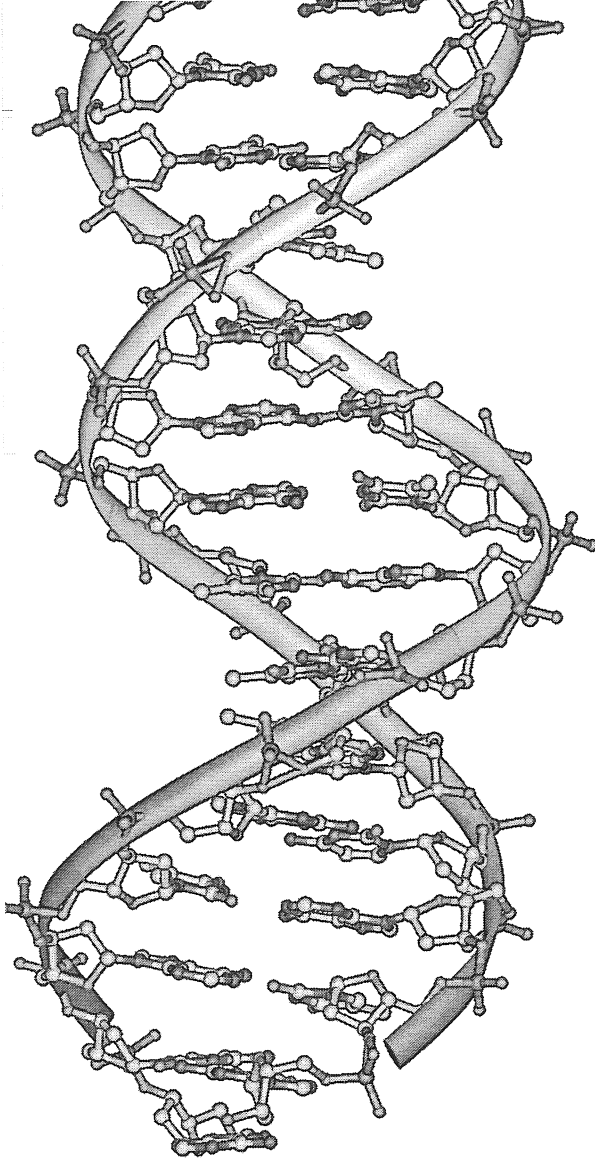


Next click on the 'what is a chromosome' link and answer the following questions.

7. How long is all of the DNA in a human cell and how does it all fit in the nucleus?

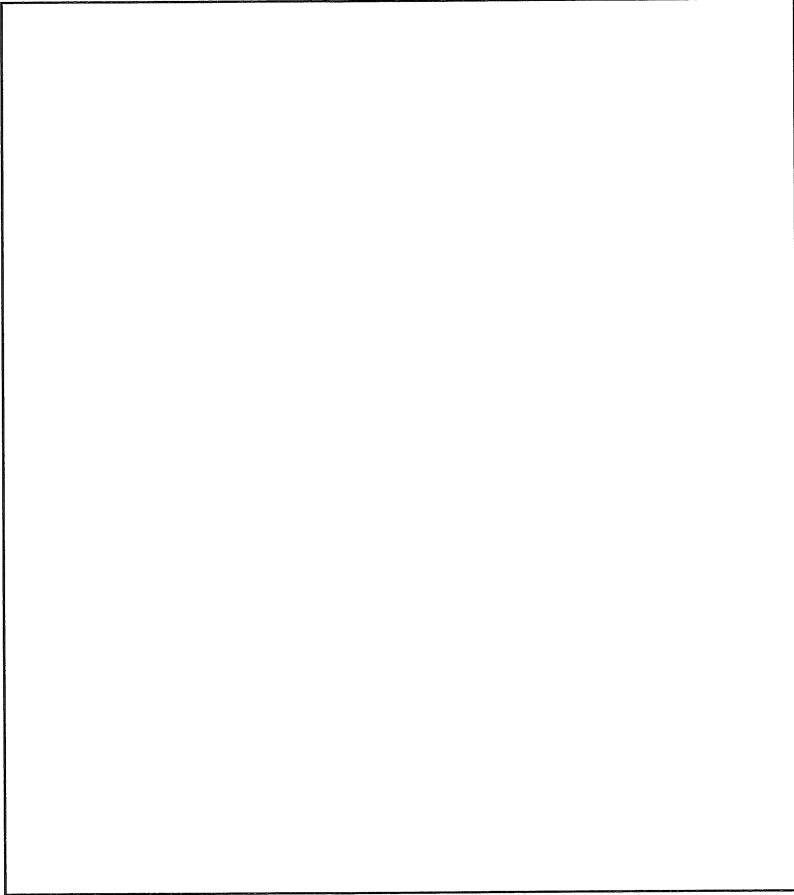
8. How many chromosomes does a human cell have and how many pairs does this make?

9. Do all animals have the same number of chromosomes? Explain.



Questions

1. How successful were you separating out some DNA? Explain. Draw a side on view of your beaker showing the DNA.



2. Take two of the steps in the procedure and explain what role they played in separating out the DNA.

3. Describe how you think this experiment could be improved.

Barcode DNA

The process of extracting and collecting DNA from a sample of human tissue is very involved. The end result that is used in law courts is a DNA Profile. This is a photograph/transparency of bands of DNA separated by a process called electrophoresis. These are similar in appearance to a 'barcode'. In this activity barcodes will be used to represent DNA profiles.

Aim – To determine the identity of a 'criminal' by matching a DNA profile found at a crime scene to those of a number of suspects.

Materials - (*Teachers refer to Teachers Guide*)

- Transparency of a DNA profile taken as evidence from a crime scene
- Worksheet of DNA profiles taken from a range of suspects

Procedure

- 1) Your teacher will give you a worksheet containing a number of DNA profiles of suspects from a crime; glue this onto the next page.
- 2) You will also be given a numbered transparency of a DNA profile taken as evidence from a crime scene.
- 3) Overlay the transparency over each of the suspects DNA profiles until you have a match.
- 4) Record the letter of the evidence profile and the corresponding number of the suspect profile in the results table.
- 5) Return the transparency to the front bench and repeat for another seven.

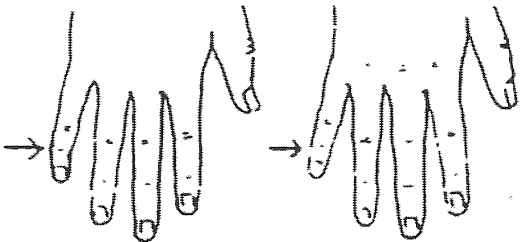
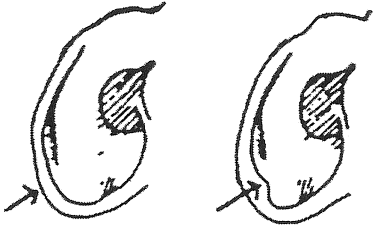
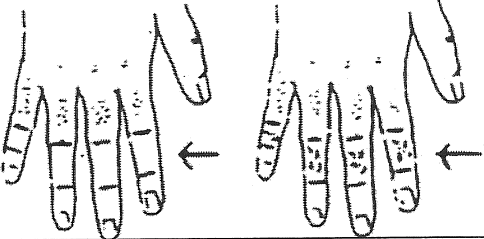
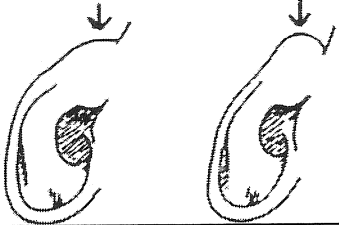
Results

Evidence Profile Number	Suspect Profile Name

Genetically determined variability in humans

DNA carries the genetic information that determines an individual's features. Except for identical twins, it is highly unlikely that any two people would have the exactly the same combination of genetically determined traits.

Use the following information to determine and record your characteristics for the twenty traits below, in the table that your teacher will give to you (*Teachers see Teachers Guide*), and then compare your traits with others in the class to assess who is genetically most similar to you.

	1. Bent little finger Hold your hands before your face with the palms toward you. Place the two little fingers side by side and press them together. The end segments of the little fingers may be straight or turned towards the ring finger. Bent (B) is dominant over (b) straight.
	2. Darwin's Point (tubercle) Is a projection that varies in size on the upper inner rim of the ear. The tubercle (P) is a dominant characteristic. Its absence (p) is recessive.
Can you bend your thumb back to form an arc shape, if you can you have distal hyper extensibility or 'hitch hikers thumb'. Hitch hikers thumb (D) is dominant over normal thumb (d).	3. Thumb hyperextension
	4. Mid-digital hair Do you have hair on the back of the middle segments of the fingers, the presence of hair on one or more middle segments of the fingers is dominant (H). Lack of mid-digital hair is a recessive trait (h).
	5. Ear-lobes Ear-lobes that hang free (E) are dominant to attached ear-lobes (e).
6. Earwax There are two forms of earwax that occur in the auditory canal. Stick your finger in your ear; the dominant type of ear wax (W) is sticky and yellow to brown in colour. The recessive form (w) is grey and crumbly.	
7. Shape of nose A nose that is convex (has a hump) and is prominent (large) is dominant (N) to a straight and moderate nose (n).	

Genetic family features

You are a combination of your fathers and mothers DNA, who in turn are a combination of their mothers and fathers (your grandparents DNA). Your teacher may be able to tell you of situations on parents' nights where although they have not met them before they can identify your parents due to a strong family resemblance.

In a similar fashion to the previous activity choose one or more family members who are genetically related to you and complete the table below.

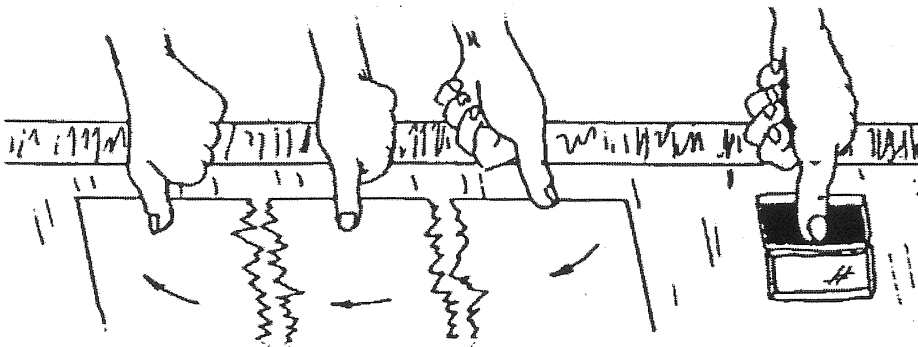
Trait	You	?	?	?
Bent little finger				
Darwin's Point				
Thumb hyperextension				
Mid-digital hair				
Ear-lobes				
Earwax				
Shape of nose				
Dimples				
Eye colour				
Length of eyelashes				
Widow's peak				
Handedness				
Hair form				
Tongue rolling				
Hair whorl				
Finger nails				
Freckles				
Face shape				

Looking at your own prints

Aim – To make copies of your own fingerprints.

Materials

- Stamp pad (artists water soluble ink)
- Paper towel
- Soap



Procedure

- 1) Place your inkpad on the table.
 - 2) Using one finger at a time roll your finger from left to right on the ink pad and then do the same on the spaces provided below.
- HINT** - Try some practice prints on some scrap paper first!

LEFT THUMB

LEFT INDEX

LEFT MIDDLE

LEFT RING

RIGHT INDEX

RIGHT MIDDLE

RIGHT RING

RIGHT THUMB

Measurement

In most experiments in 'Human Biological Sciences' you will be required to make accurate measurements of physical quantities, for example, time, length, volume and weight. To make these measurements accurate it will be necessary to use scientific equipment such as electronic timers, rulers, measuring cylinders and scales.

It is important that you know how to use this equipment correctly and in a safe manner. Use a ruler and an electronic balance to measure and calculate the following.

Note – be as accurate as you can.

Length (L) of this workbook = _____

Width (W) of this workbook (ignore the black binder) = _____

Height (H) of this workbook = _____

Weight (G) of this book = _____

Volume (V) of this book = $L \times W \times H$ = _____

Density of this book = G / V = _____

Calculation of averages

To calculate averages add together the original data from the trials and divide by the number of trials. Note your averages should be rounded off.

Example

For example if a student wanted to work out the average weight of five rats, they would weigh each rat add them together and divide it by the number of measurements

Eg. Rat 1 – 126g, Rat 2 – 135g, Rat 3 – 119g, Rat 4 – 127g, Rat 5 – 140g,
 $126 + 135 + 119 + 127 + 140 = 647$
 No of trials = 5
 $647 / 5 = 129$
 Average = 129 grams

Notice that the average has been rounded off to the same number of decimal places as the data and that units have been used in the answer.

Practice questions

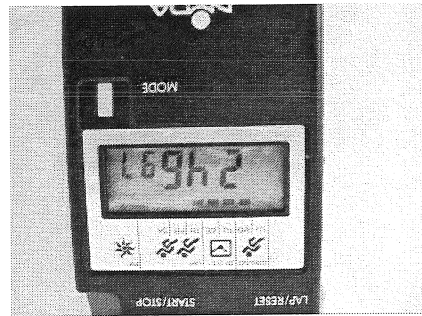
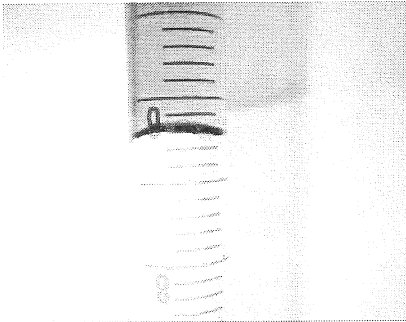
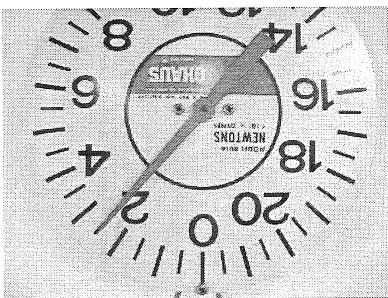
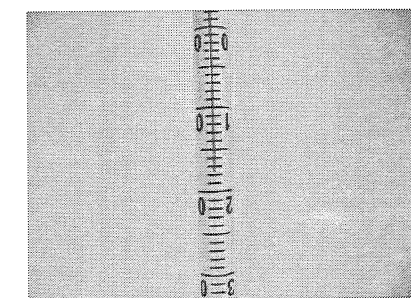
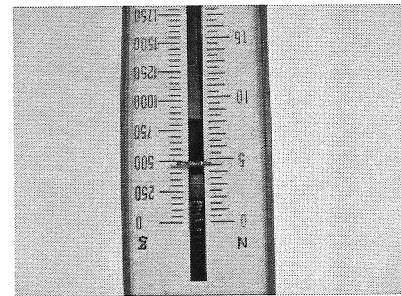
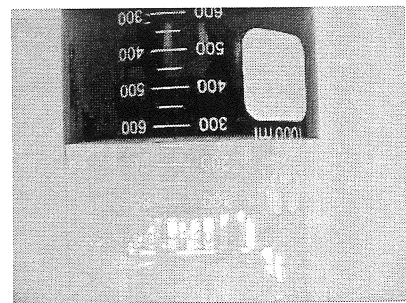
1. Four students recorded the amount of work they did around the house for a week, Renee did 8 hours, Joshua recorded 6 hours, Laura did 7 hours and Tim said 6. What is the average number of hours worked around the house for these students? Show your working.

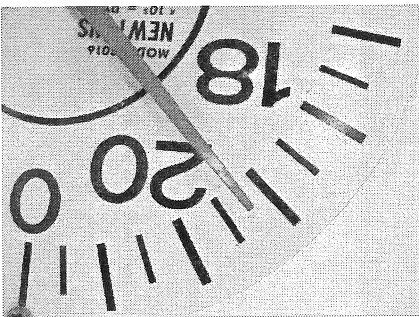
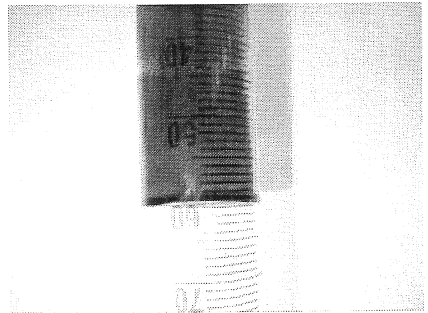
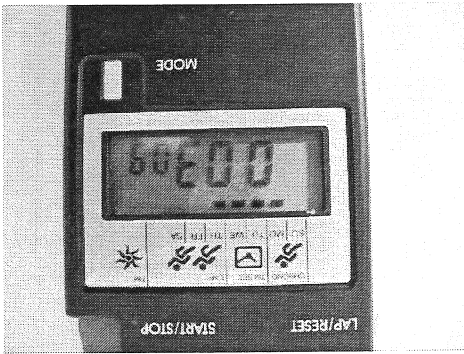
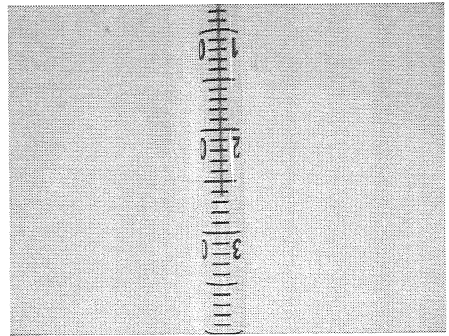
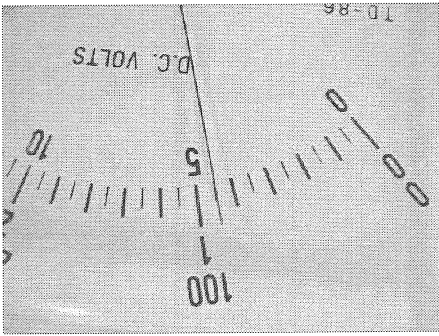
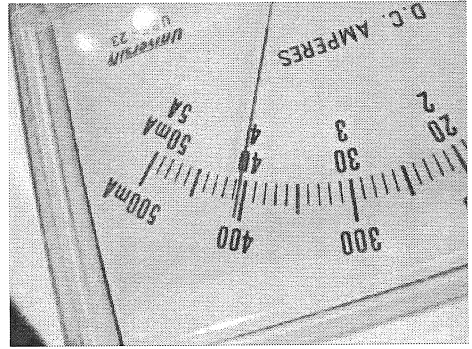
2. An enterprising Science student used the skills he had gained in 'Human Biological Science' to set up a hydroponic garden. He planted 10 tomato seeds and recorded the height of each plant after 5 weeks. His measurements in centimetres were; 36, 45, 52, 39, 41, 30, 45, 50, 38 & 42. Calculate the average height of his crop of plants. Show your working.

3. When his crop had flowered and produced their fruit he picked the fruit and weighed how many tomatoes was produced by each plant. His measurements in kilograms were; 3.2, 5.1, 4.2, 4.0, 4.9, 3.8, 4.0, 3.9, 4.7 & 4.5. Calculate the average weight of tomatoes produced by each plant. Show your working.

Scales

When using scientific equipment to make quantitative measurements it is important that you can accurately read the scales. For each of the figures following write down the numerical measurement, some will have more than one possible reading (do not worry about units in these practice examples).





- Some hypotheses in the “If / Then” format would be;
- “If the concentration of carbon dioxide in the blood decreases then the heart rate will decrease.”
 - “If the temperature of the body rises above normal then the rate of sweating will increase.”
 - “If the amount of light entering the pupil increases then the iris will contract to reduce the size of the pupil.”

3. Identifying the independent variable

Anything factor that can be changed or varied during an experiment or that can affect the outcome of an environment it termed a variable. When designing an experiment only one variable is selected to be varied in the experiment. The variable selected is called the independent variable. Because the investigator can alter this variable it is sometimes referred to as the manipulated or experimental variable.

In general terms it is the variable that the investigator deliberately changes in the experiment. In this experiment the independent variable is the type of mince, low fat or normal.

4. Identifying the dependent variable

The measurable change as a result of the independent variable being altered is called the dependent or responding variable.

In general terms it is the variable that is measured in the investigation and in this experiment it is the amount of fat.

5. Identifying the controlled variables

In order to make the experiment a fair test all other variables have to be kept the same during the experiment. These variables are called controlled variables. Careful thought needs to be given to all the possible factors that could affect the experiment before you start.

The control variables in this investigation are:

- a) weight of mince
- b) amount of water
- c) boiling time
- d) weighing balance used.

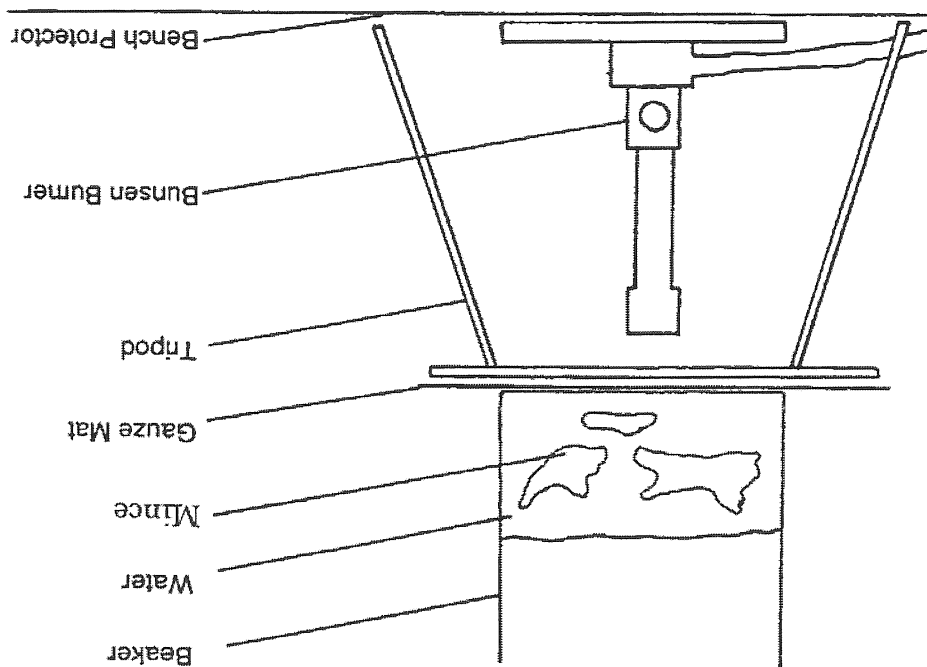
6. Planning a valid experiment

Generally experiments in ‘Human Biological Sciences’ will have two ‘set-ups’ or groups operating. The set-up/group that is ‘normal’ is termed the ‘control set-up/group’. The set-up/group that is exactly the same as the control except one difference has been made (e.g. the independent variable) is termed the ‘experimental set-up/group’.

Errors in an experiment can occur due to errors making measurements, equipment problems and biological variation. To help make the conclusion of an experiment more reliable that are a number of procedures that can utilised to reduce the EFFECT of these errors (Note – you don’t reduce the chance of or the errors themselves).

9. Diagram of experimental apparatus

You should draw a diagram where possible to help explain the method. Diagrams should be in pencil, 2-dimensional (side view), labelled, large and clear. Here is an example for you.



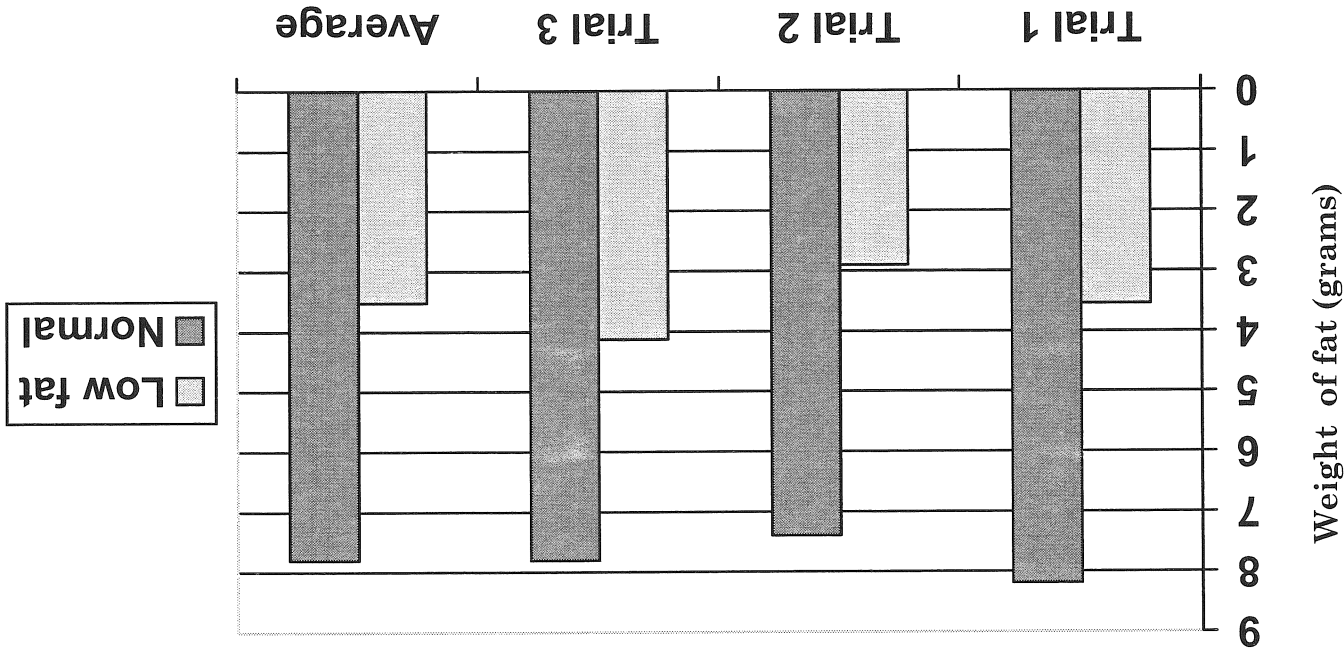
10. Experimental errors

When designing and completing experiments there are many potential sources of error. The errors that you are most likely to make are measurement errors.

In most experiments you have to make accurate measurements of physical quantities such as lengths and weights etc, or recordings of time intervals. It is important to use scientific equipment like timers, rulers and scales to make these measurements, and to know how to use this equipment correctly.

You also need to think about any factors that may influence your measurement and take appropriate actions to make sure that they do not. In this experiment for example it is important to use scales (or balances) to measure the weight of mince and fat plus a stopwatch to measure the time.

FAT CONTENT OF 100 g OF MINCE



13. Writing a conclusion

The conclusion is generally towards the end of an experimental report. Conclusions should be brief and not introduce any new ideas or information not dealt with earlier. Your conclusions should be presented in a series of separate statements.

Conclusions state:

- Whether or not the data supports or not supports the hypothesis (backed up with results).
- a repeat of the hypothesis.
- a general statement concerning the results.
- Any inferences that can be drawn from the data.

An example of a good conclusion would be:

- Our results support the hypothesis "That low fat mince has less fat than normal mince". The average fat content for normal mince was 7.80% and low fat mince 3.52%. This meant that normal mince in this test had over double the fat of low fat mince. So maybe the butcher's customers should not have been complaining. An interesting point, however, is that low fat mince still has quite a lot of fat, 3.52%.

Results

[illegible][illegible]

Questions

1. Write a hypothesis for this experiment.

2. Which group was the control group? Which group was the experimental group?

3. What was the sample size?

4. What is the independent variable (the one you changed)?

5. What is the dependent variable (the one you measured)?

6. List four controlled variables (the ones you keep the same).

7. What would be the term for you doing the same experiment again next week?

8. What would be the term for another class doing the same experiment next week?

9. Compare the graph of the non-caffeine group to that of the caffeine group.

10. What effect does caffeine appear to have on pulse rate?

11. Cola contains 13 milligrams of caffeine per 100 mLs. How many milligrams of caffeine would there be in a 375 mL can of cola?

12. Caffeine does not have a noticeable taste. Why then do you think manufacturers of cool drinks add caffeine to their products?

13. Determine the percentage change in your pulse rate from its resting level to the highest point it reached in the 10 minute period. Show your working.